

Hospital Chatbot for Color Coded Clinical Pathways

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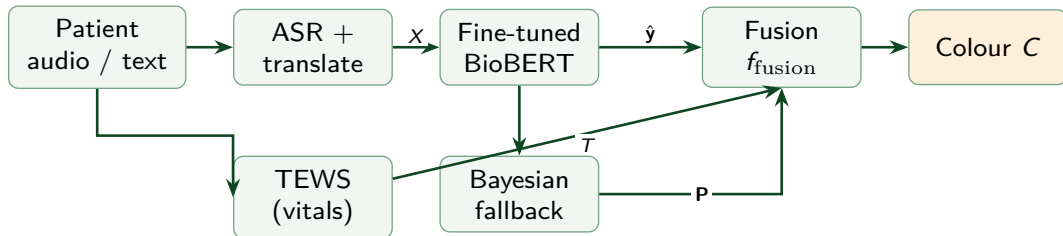
Outline

- 1 Clinical Foundations
- 2 Patient Input Pipeline
- 3 BioBERT NLP Layer
- 4 Fusion and Parallel Paths
- 5 Mathematics Deep Dives
- 6 End-to-End Worked Example
- 7 Clinical Protocols and Demo

System overview

Curatio is an AI-assisted hospital triage chatbot that assigns a **single SATS colour** C from:

- what the patient **says** (text or voice),
- what vitals imply via **TEWS**, and
- **Bayesian** reasoning when data are incomplete or the model is uncertain.



Session state: $\mathbf{S}_s = (X, \mathbf{D}, T, \mathbf{P}, C, \text{flags})$

Session mathematics (all layers)

$$\mathbf{D} = f_{\text{NLP}}(X), \quad T = f_{\text{TEWS}}(\mathbf{v}), \quad \mathbf{P} = f_{\text{Bayes}}(E), \quad C = f_{\text{fusion}}(\mathbf{D}, T, \mathbf{P})$$

Symbols

X	English chief complaint
$\hat{\mathbf{y}}$	5-class acuity probabilities
T	TEWS total score
E	Evidence vector (entities + vitals)
C	Final SATS colour
τ	Confidence threshold (≈ 0.85)

In plain English

Each layer produces an **auditable** signal. Fusion is a **safety checklist**, not an average — the most urgent credible signal wins.

What is triage?

Triage sorts patients by clinical urgency so the sickest are seen first.

Curatio uses the **South African Triage Scale (SATS)**, validated at **Komfo Anokye Teaching Hospital (KATH)**, Kumasi, Ghana.



In plain English

Think of triage as a traffic light for the ED: each colour tells staff how quickly a patient must be seen. The system **supports** clinical staff — it does not replace their judgment.

SATS colour codes

Colour	Target time	Clinical meaning
triageRed Red	0 min	Immediate resuscitation — life-threatening
triageOrange Orange	< 10 min	Very urgent — high dependency
triageYellow Yellow	< 60 min	Urgent — needs physician review
triageGreen Green	< 4 hours	Non-urgent — routine care
triageBlue Blue	< 2 hours	Deceased on arrival — dignity protocol

Each colour triggers a **clinical pathway** with strict time limits and department routing.

Colour → pathways → departments (Ghana)

Colour	Protocol	Destination	System action
triageRed Red	RED	Resuscitation Room	Code Red; bypass registration/payment
triageOrange Orange	ORANGE	Majors (ED)	10-min countdown; cardiology alert
triageYellow Yellow	YELLOW	Majors (ED)	Standing orders (e.g. malaria test)
triageGreen Green	GREEN	Minors / OPD / Polyclinic	Divert non-urgent from ER
triageBlue Blue	Mortuary	—	Silent dignity protocol; notify social workers

In plain English

Curatio routes patients automatically by assigned colour — maintaining **acuity continuity** across departments (the hospital does not “forget” how sick someone is).

Step 1: How data enters the system

Session start: patient describes symptoms via the Curatio chatbot.

Two input modes

- **Typed text** — stored directly as raw complaint
- **Voice note** — Twi / local Ghanaian language

5-step pipeline

- 1 Patient input (this slide)
- 2 Text example
- 3 Speech & translation
- 4 Fine-tuned BioBERT
- 5 Fusion → colour C

TEWS and Bayesian join at Step 5.

In plain English

Whether the patient types or speaks, the goal is the same: produce English text X for the NLP layer.

Step 2: Example text input

Typed complaint is stored as `chief_complaint_raw` and becomes text X (max 128 tokens).

Worked example

Patient ID: TG-UXRGA9UCO

Complaint: *"thunderclap headache, worsening with movement"*

Nurse ground truth: Acuity Level 2 — **triageOrange**Orange (highly urgent)

Voice input skips this direct path and goes to Step 3 (ASR + translation) before BioBERT.

In plain English

Real hospital phrasing — not textbook English — is exactly what fine-tuning teaches BioBERT to handle.

Step 3: Speech and translation

$$X = g_{\text{trans}}(g_{\text{ASR}}(\text{audio}))$$

- g_{ASR} — automatic speech recognition (Twi $\rightarrow$ transcript)
- g_{trans} — medical translation (Twi $\rightarrow$ English)
- Typed English from Step 2 bypasses both functions

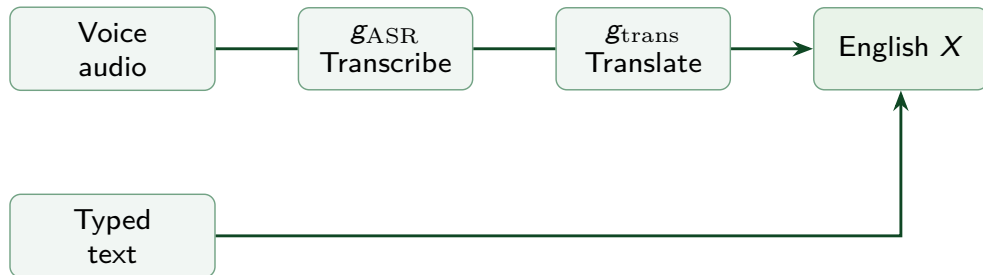
Worked example

Patient TG-001 (Twi voice):

- 1 Audio: *"Me koma mu ye me ya, me home ye me"*
- 2 ASR: *Me koma mu ye me ya*
- 3 Translation: *"I have chest pain, I feel unwell"*
- 4 $X \rightarrow$ BioBERT in Step 4

In plain English

Step 3: Pipeline diagram



Both paths converge on the same X before the BioBERT layer.

Step 4: Fine-tuned BioBERT overview

English text X passes through our fine-tuned BioBERT model $\rightarrow$ acuity prediction $\hat{y}$.

$$X \xrightarrow{\tau} (t_i) \xrightarrow{E} H^{(0)} \xrightarrow{L \times} H^{(L)} \xrightarrow{\text{head}} \hat{y}$$

Predicted acuity	Maps to SATS
Level 1–2	Red / Orange (highly urgent)
Level 3	Yellow (moderate)
Level 4–5	Green (non-urgent / routine)

In plain English

Fine-tuning taught BioBERT our hospital's phrasing. It answers "how urgent does this sound?" — that answer goes to fusion with vitals and Bayesian when needed.

Step 4: Softmax classification

$$\hat{\mathbf{y}} = \text{softmax}(W\mathbf{h}_{[\text{CLS}]} + \mathbf{b}), \quad \hat{y}_c = P(\text{acuity} = c \mid X)$$

Worked example

Chief complaint: thunderclap headache, worsening with movement

Level	Meaning	Probability
1	Most urgent	1%
2	Highly urgent	94%
3	Moderate	3%
4–5	Routine	2%

Prediction = Level 2 → maps to **triageOrange**Orange

In plain English

Softmax converts raw scores into percentages over 5 acuity levels — highest wins.

Step 4: Confidence gate

confidence = $\max_c \hat{y}_c$, if confidence $< \tau \Rightarrow$ flag for Bayesian fallback

where $\tau \approx 0.85$.

Worked example

High confidence (passes): $\max(\hat{y}) = 0.94 > \tau = 0.85 \rightarrow$ trust BioBERT.

Borderline (flagged): Patient TG-7OKLDLKAQ — glaucoma complaint:
Level 2: 85.4%, Level 3: 14.1% $\Rightarrow$ confidence 0.854 — fusion may call Bayesian fallback.

In plain English

When the model is unsure, we do not silently trust it — fusion escalates to Bayesian reasoning.

BioBERT: inside the model — embeddings

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(t_i)$$

Each token gets three vectors summed: word identity, position in sentence, segment (question vs. answer).

Worked example

Input: “crushing chest pain radiating to left arm”

Tokens: [crush, ##ing, chest, pain, radiate, ##ing, left, arm]

Each token → 768-dim vector after embedding layer.

BioBERT: self-attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right) V$$

Worked example

Query token “crushing” attends most strongly to:

- “crushing” — 0.38
- “chest” — 0.35
- “pain” — 0.22

Clinical co-occurrence drives urgency detection.

BioBERT: transformer layers and [CLS]

$$Z^{(\ell)} = \text{LayerNorm}\left(H^{(\ell-1)} + \text{MultiHead}(H^{(\ell-1)})\right)$$

$$H^{(\ell)} = \text{LayerNorm}\left(Z^{(\ell)} + \text{FFN}(Z^{(\ell)})\right)$$

After $L = 12$ layers:

$$h_{[\text{CLS}]} = H_{1,:}^{(L)} \in \mathbb{R}^{768} \quad \hat{\mathbf{y}} = \text{softmax}(Wh_{[\text{CLS}]} + \mathbf{b})$$

Worked example

“crushing chest pain” → Level 2: 91%, Level 3: 6%, others: 3%

Confidence 0.91 → **triageOrange**Orange at fusion.

BioBERT: how it was trained — pre-training

$$\mathcal{L}_{\text{MLM}} = - \sum_{i \in \mathcal{M}} \log P(t_i | H^{(L)}; \theta)$$

Masked Language Modelling on PubMed medical English teaches BioBERT clinical vocabulary.

Worked example

“The patient presented with severe _____ pain”

Masked token predicted: **chest**

Model learns co-occurrence of medical terms before ever seeing triage labels.

BioBERT: fine-tuning on hospital data

$$\mathcal{L} = - \sum_{i=1}^N \log P(y_i | X_i; \theta)$$

Minimise cross-entropy vs. nurse-assigned acuity labels on $\sim 80,000$ chief complaints. **Vitals excluded** — model learns urgency from language alone.

Worked example

Row: “contraception advice, intermittent” $\rightarrow$ nurse label Level 5

$P(\text{Level 5} | X) = 0.97 \Rightarrow \mathcal{L} += -\log(0.97) \approx 0.03$

Setting	Value
Base model	Yuvrajxms09/biobert-triage-classifier
Data merge	train.csv $\bowtie$ chief_complaints.csv ($\sim 80k$ rows)
Epochs	3 (13,500 steps)
Best eval_loss	0.001812

Data exploration and findings

Training and inference on $\sim$ **100,000** patient records from fine-tuned BioBERT CSVs.

Dataset sizes

File	Rows
train.csv	80,000
test.csv	20,000
chief_complaints.csv	100,000
triage_predictions_results.csv	100,000

Acuity distribution (train)

Level	Count	%
1	3,222	4.0%
2	13,439	16.8%
3	28,921	36.2%
4	23,020	28.8%
5	11,398	14.2%

Confidence: 99.6% of predictions at confidence 1.0; **361** rows at 0.6–0.99 (Bayesian candidates).

In plain English

Some urgent cases are overconfidently misclassified — fusion with TEWS/Bayesian mitigates this.

Example predictions from CSV

Patient ID	Complaint	Pred.	Conf.
TG-UXRGA9UCO	thunderclap headache, worsening with movement	Level 2	1.00
TG-B19DBBS2G	contraception advice, intermittent	Level 5	1.00
TG-7OKLDLKAQ	acute angle closure glaucoma in known patient	Level 2	0.854
TG-2MIQ3DLS7	acute angle closure glaucoma with nausea	Level 2	0.855

Glaucoma cases at ~ 0.85 confidence illustrate why the $\tau \approx 0.85$ gate and Bayesian fallback exist.

Step 5: Fusion → triage colour

$$C = f_{\text{fusion}}(\mathbf{D}, T, \mathbf{P}, \text{flags})$$

Fusion combines BioBERT ($\hat{\mathbf{y}}$), TEWS (T), and Bayesian ($\mathbf{P}$) into one SATS colour C .

Worked example

	Input	Value	Suggests
Chest pain patient at KATH:	BioBERT $\hat{\mathbf{y}}$	Level 2, 94% conf	triageOrange Orange
	TEWS T	4	triageYellow Yellow
	Bayesian P	$P(\text{Orange} \mid E) = 0.89$	triageOrange Orange
$\Rightarrow C = \text{triageOrangeOrange}$ (symptoms beat mild vitals)			

In plain English

f_{fusion} is a **safety checklist**, not an average — always picks the more urgent credible signal.

Parallel path: vitals $\rightarrow$ TEWS

Six vitals summed with weights $w_k = 1$ — deterministic, no AI.

$$T = f_{\text{TEWS}}(\mathbf{v}) = \sum_{k=1}^6 w_k f_k(v_k), \quad w_k = 1$$

$$C_{\text{TEWS}}(T) = \begin{cases} \text{Red} & T > 7 \\ \text{Orange} & 5 \leq T \leq 6 \\ \text{Yellow} & 3 \leq T \leq 4 \\ \text{Green} & T \leq 2 \end{cases}$$

Worked example

HR = 125 bpm $\rightarrow f_1(125) = 2$; RR = 26/min $\rightarrow f_2(26) = 2$; $f_{3:6} = 0$

$T = 4 \Rightarrow C_{\text{TEWS}} = \text{triageYellow Yellow}$

Fusion may still upgrade to Orange if language/Bayesian say so.

Parallel path: Bayesian fallback

Bayesian reasoning runs when:

- vitals are missing (partial TEWS),
- BioBERT confidence $< \tau \approx 0.85$, or
- language and vitals strongly disagree.

$$P(C_k | E) = \frac{P(E | C_k) P(C_k)}{\sum_j P(E | C_j) P(C_j)}, \quad C_{\text{Bayes}} = \arg \max_k P(C_k | E)$$

Worked example

Chest pain, partial vitals at KATH:

Colour	Prior $P(C_k)$	$P(E C_k)$	$P(C_k E)$
Red	5%	Low	8%
Orange	15%	High	89%
Yellow	30%	Medium	2%
Green	50%	Very low	1%

Disease override rule

$$P(D | S) = \frac{P(S | D) P(D)}{P(S)}$$

If $P(D | S) > \tau_B$ then $C_{\text{Bayes}} = \text{Red (or Orange per protocol)}$

Worked example

Symptoms: crushing central chest pain, radiating to left arm

$$P(\text{MI} | S) \approx 0.82 > \tau_B = 0.75$$

$\Rightarrow$ upgrade to Red/Orange even if $T = 2$ (Green vitals alone)

In plain English

Serious disease suspicion can override mild vital scores — clinical safety first.

TEWS: heart rate scoring $f_1(HR)$

$$f_1(v_1) = \begin{cases} 3 & \text{if } HR \geq 130 \\ 2 & \text{if } 111 \leq HR \leq 129 \\ 0 & \text{if } 51 \leq HR \leq 100 \\ 2 & \text{if } HR \leq 40 \\ 1 & \text{otherwise (borderline)} \end{cases}$$

Worked example

HR = 125 bpm:125 $\geq$ 130? No. 111 $\leq$ 125 $\leq$ 129? **Yes** $\Rightarrow f_1(125) = 2$ points.

Every score traceable to a SATS table row — auditable for clinical review.

TEWS: respiratory rate scoring $f_2(RR)$

$$f_2(v_2) = \begin{cases} 3 & \text{if } RR \geq 30 \\ 2 & \text{if } 21 \leq RR \leq 29 \\ 0 & \text{if } 9 \leq RR \leq 14 \\ 1 & \text{otherwise} \end{cases}$$

Worked example

RR = 26/min: $26 \geq 30$? No. $21 \leq 26 \leq 29$? **Yes** $\Rightarrow f_2(26) = 2$ points.

TEWS: partial scoring when vitals missing

$$T_{\text{partial}} = \sum_{k \in \mathcal{K}_{\text{obs}}} w_k f_k(v_k), \quad \mathcal{K}_{\text{obs}} = \{k : v_k \text{ measured}\}$$

Worked example

Only HR and RR measured; temp, AVPU, mobility, trauma unknown.

Sum only observed f_k ; set `tews_incomplete = true`.

Fusion **cannot** assign Green on partial data — escalates to Bayesian.

TEWS total	Colour	Ghana routing
$T > 7$	Red	Resuscitation Room
$5 \leq T \leq 6$	Orange	Majors ED
$3 \leq T \leq 4$	Yellow	Majors ED
$T \leq 2$	Green	Minors / OPD

TEWS worked example

$$f_1(125) = 2, \quad f_2(26) = 2, \quad f_{3:6} = 0 \Rightarrow T = 4 \Rightarrow C_{\text{TEWS}} = \text{Yellow}$$

Vital	Value	f_k	Points
HR	125 bpm	$f_1(125)$	2
RR	26/min	$f_2(26)$	2
Mobility, Temp, AVPU, trauma	Normal	0 each	0
Total T			4

Bayesian: evidence vector E

$$E = [\text{entity}_1, \dots, \text{entity}_r, v_{k_1}, \dots, v_{k_s}]$$

Combines NLP-extracted clinical entities with whatever vitals were measured.

Worked example

NLP entities: chest_pain, radiating_pain

Vitals: HR = 125, RR = 26 (temperature missing)

$E = [\text{chest_pain}, \text{radiating_pain}, \text{HR}=125, \text{RR}=26]$

Bayesian: posterior and decision

$$P(C_k | E) = \frac{P(E | C_k) P(C_k)}{\sum_j P(E | C_j) P(C_j)}, \quad C_{\text{Bayes}} = \arg \max_k P(C_k | E)$$

- **Prior** $P(C_k)$ — baseline urgency from hospital epidemiology
- **Likelihood** $P(E | C_k)$ — how well symptoms fit each colour band
- **Posterior** — updated belief after seeing evidence E

In plain English

When TEWS is blind (missing vitals), Bayesian uses whatever evidence exists to estimate risk.

Fusion: urgency ordering

$$\text{ord}(\text{Red}) = 4 > \text{ord}(\text{Orange}) = 3 > \text{ord}(\text{Yellow}) = 2 > \text{ord}(\text{Green}) = 1$$

Worked example

Language suggests Orange ($\text{ord} = 3$), TEWS suggests Yellow ($\text{ord} = 2$).

$\max(3, 2) = 3 \Rightarrow C = \text{triageOrangeOrange}$.

Language wins — chest pain cannot wait for mild vital derangement alone.

Fusion: safety rule

$$\text{ord}(C) \geq \max\{\text{ord}(C_{\text{disc}}), \text{ord}(C_{\text{TEWS}}), \text{ord}(C_{\text{Bayes}})\}$$

Final colour C is **never less urgent** than any layer suggested.

Worked example

TEWS alone $\rightarrow$ Yellow; BioBERT + Bayesian $\rightarrow$ Orange.

Safety rule: $\text{ord}(C) \geq 3 \Rightarrow C = \text{triageOrangeOrange}$.

Fusion: Algorithm 1 checklist

- ① Life-threatening words? $\rightarrow$ **triageRedRed**
- ② Very urgent words (chest pain)? $\rightarrow$ **triageOrangeOrange**
- ③ $T > 7$? Red | $5 \leq T \leq 6$? Orange | $3 \leq T \leq 4$? Yellow
- ④ Missing vitals? $\rightarrow \arg \max_k P(C_k | E)$
- ⑤ Serious disease? $\rightarrow$ Red / Orange override
- ⑥ Else $\rightarrow$ **triageGreenGreen**

Language (D)	TEWS (T)	Final C
$d_{\text{chest}} > \tau$ (Orange)	$T = 4$ (Yellow)	triageOrangeOrange
Low D	$T \geq 7$ (Red)	triageRedRed
Ambiguous D	Incomplete vitals	$\arg \max P(C_k E)$

Design note: spec vs. implementation

Mathematical spec

Discriminator vector **D** with SATS pattern scores

$$d_j = \sigma(w_j^\top h_{[\text{CLS}]} + b_j)$$

Fine-tuned model

5-class acuity classifier on

`chief_complaint_raw` $\rightarrow$ mapped to SATS colours for fusion

Both feed fusion as a language-layer urgency signal. Confidence $< \tau$ triggers Bayesian regardless of architecture detail.

In plain English

The web presentation documents this gap transparently — mathematics describes the intended safety architecture; BioBERT provides the deployed NLP signal.

Kumasi chest-pain case: all layers

Scenario: Patient at KATH with central crushing chest pain. All three layers evaluated before fusion.

BioBERT

“central crushing chest pain,
radiating to left arm”

$$d_{\text{chest}} > \tau$$

⇒ Orange language signal

TEWS

HR = 125, RR = 26

$$T = 4$$

⇒ Yellow alone

Bayesian

$$P(\text{Orange} \mid E) \approx 0.89$$

⇒ Orange posterior

Worked example: fusion resolution

$$f_1(125) = 2, \quad f_2(26) = 2, \quad f_{3:6} = 0 \Rightarrow T = 4 \Rightarrow C_{\text{TEWS}} = \text{Yellow}$$

$$\underbrace{d_{\text{chest}} > \tau}_{\text{language (Orange)}} + \underbrace{T = 4}_{\text{TEWS alone = Yellow}} + \underbrace{P(\text{Orange} \mid E) \approx 0.89}_{\text{Bayesian (Orange)}} \Rightarrow C = \text{Orange}$$

Worked example

Final routing at KATH:

- $C = \text{triage}\text{Orange}\text{Orange}$
- Destination: Majors (ED)
- Cardiology alert triggered
- 10-minute countdown starts

In plain English

Resuscitation protocol (**triageRedRed**)

Target response time: 0 minutes (immediate)

① Secure airway & vitals

- Airway patency; continuous cardiac, SpO₂, NIBP monitoring

② Establish access

- Two large-bore IVs or IO access
- Stat labs: VBG, lactate, cardiac enzymes

③ Targeted intervention

- ACLS / ATLS protocols as indicated
- Defibrillation, epinephrine, massive transfusion, etc.

In plain English

Red bypasses registration and payment — Code Red activated immediately upon fusion assignment.

Curatio interactive presentation (built deliverable)

Web app: `curatio/client` — React + Vite presentation at `/dashboard`

17 slides covering:

- SATS intro & Ghana pathways
- 5-step input pipeline
- BioBERT inference & training
- Data exploration (80k CSV stats)
- TEWS & Bayesian parallel paths
- Fusion mathematics
- End-to-end worked example
- Resuscitation protocol

Interactive features:

- KaTeX formula rendering
- Flip cards: formula $\leftrightarrow$ worked example
- Collapsible sidebar navigation
- Arrow keys: $\longleftrightarrow$ slides, $\updownarrow$ scroll
- Fullscreen mode with zoom
- Pre-computed stats from fine-tuned BioBERT CSVs

`npm run dev` in `curatio/client` $\rightarrow$ open `/dashboard`

Project deliverables summary

Component	Description
Fine-tuned BioBERT	5-class acuity on chief complaints; eval_loss 0.001812
CSV datasets	100k records: train, test, complaints, predictions
Mathematics spec	TEWS piecewise, Bayesian posterior, fusion safety rules
Curatio web UI	Interactive presentation mirroring this Beamer deck
Ghana routing	SATS colours → KATH departments and protocols

This Beamer deck (`slides_saturday/`) is the static companion to the live web demo.

Summary

- ① **Clinical context:** SATS colours map to Ghana hospital departments at KATH.
- ② **5-step pipeline:** Input $\rightarrow$ text/voice $\rightarrow$ BioBERT $\rightarrow$ fusion with TEWS & Bayesian.
- ③ **BioBERT:** Fine-tuned on $\sim 80k$ chief complaints; 5-class acuity softmax + confidence gate.
- ④ **TEWS:** Deterministic vital scoring; piecewise $f_1 \dots f_6$.
- ⑤ **Bayesian:** Posterior $P(C_k | E)$ when vitals incomplete or NLP uncertain.
- ⑥ **Fusion:** $\text{ord}(C) \geq \max\{\text{ord}(C_{\text{NLP}}), \text{ord}(C_{\text{TEWS}}), \text{ord}(C_{\text{Bayes}})\}$.

Interactive demo: `curatio/client` React presentation at `/dashboard`

Thank you

Questions?